



PHILIPPINE STATE COLLEGE OF AERONAUTICS



INSTITUTE OF COMPUTER STUDIES

Name of Institute



BS INFORMATION TECHNOLOGY

with specialization in

Aviation Information Technology

Name of Program



CAPSTONE PROJECT 2

MIDTERM MODULES



by:

ASSOC. PROF. MARY ANN F. ABALLIAR-VISTA, MEAM



DISCLOSURE



This module is intended for the exclusive use of BSIT-AIT Students taking up Capstone Project 2 at the Philippine State College of Aeronautics – Institute of Computer Studies.

No part of this material may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior written permission from the Author.





Table of Contents

Honesty and Safety Clauses	5
MIDTERM	6
MODULE 1	6
Course Learning Outcome (CLO).....	6
Module Learning Outcome (MLO 7).....	6
Topic Learning Outcomes (TLO)	6
Learning Objectives	6
Required Resources	7
Module Overview	7
Introduction	8
Lesson 1: Introduction to Chapter V	8
Lesson 2: Writing the Summary of Findings.....	12
Lesson 3: Formulating Research Conclusions	16
Lesson 4: Writing Evidence-Based Recommendations	22
Lesson 5: Common Errors in Chapter V.....	28
References	33
Guided Learning Activity / Performance Task	35
Learning Outcomes Addressed	35
Activity Description	35
Instructions:.....	35
Part I – Summary of Findings (30 points)	35
Part II – Research Conclusions (30 points)	36
Part III – Evidence-Based Recommendations (20 points)	36
Part IV – Chapter V Self-Evaluation Checklist (20 points)	36
Expected Output	37
Assessment Rubric	37
MODULE 2	39
Research Day: Consultation and Revision Workshop	39



Course Learning Outcomes (CLO)	39
Module Learning Outcome (MLO 8).....	39
Topic Learning Outcomes (TLO)	39
Learning Objectives	39
Required Resources	40
Module Overview	40
Introduction	40
Learning Content.....	41
Consultation Checklist	45
References:.....	45
Guided Learning Activity.....	47
Formative Assessment.....	48
Summative Assessment: Research Day Consultation Portfolio	48
Rubric	49
MODULE 3	50
Preparing for the Mock Final Defense	50
Course Learning Outcomes (CLO)	50
Module Learning Outcome (MLO 9).....	50
Topic Learning Outcomes (TLO)	50
Learning Objectives	51
Preparing for the Mock Final Defense	51
References:.....	53
Guided Learning Activity.....	55
Formative Assessment.....	55
Summative Assessment.....	55
MODULE 4	57
Final Defense Presentation and Technical Justification.....	57
Course Learning Outcomes (CLO)	57
Module Learning Outcome (MLO 10).....	57
Topic Learning Outcomes (TLO)	57
Learning Objectives	58



Required Resources	58
Module Overview	58
Introduction	59
References	60
Institutional References	60
Guided Learning Activity.....	62
Summative Assessment.....	62
Rubrics	63
MODULE 5	65
Post-Defense Revision and Compliance Preparation	65
Course Learning Outcomes (CLO)	65
Module Learning Outcome (MLO 11).....	65
Topic Learning Outcomes (TLO)	65
Learning Objectives	65
Required Resources	66
Module Overview	66
Introduction	66
References	68
Institutional References	68
Guided Learning Activity.....	69
Formative Assessment.....	69
Summative Assessment	70
Rubrics	70
MIDTERM EXAMINATION	72
Week: 12 Time Allotment: 3 Hours.....	72
Midterm Examination.....	72
Examination Components	72
1. Revised Capstone Manuscript.....	72
2. Updated System	72
3. Updated Revision Matrix.....	72
4. Compliance Package.....	72



References	73
Examination Procedures	74
Expected Outputs.....	74
Assessment Method.....	74
Performance Descriptors	75



Honesty and Safety Clauses

Honesty Clause

Students must submit original and properly cited work. Plagiarism, data fabrication, unauthorized copying, and misrepresentation of outputs are prohibited. Any use of AI tools must be reviewed, verified, and disclosed when required.

Safety Clause

Students must observe proper safety, data privacy, information security, and institutional protocols during system development, testing, deployment, and turnover. Any technical, security, or safety concern must be reported immediately to the adviser or authorized personnel.



MIDTERM

MODULE 1

Writing Chapter V: Summary of Findings, Conclusions, and Recommendations

Week: 7 | Time Allotment: 3 Hours

Course Learning Outcome (CLO)

CLO 4. Prepare a complete, technically sound, and institutionally compliant capstone manuscript by synthesizing research findings, drawing evidence-based conclusions, and formulating practical recommendations based on the completed study.

Module Learning Outcome (MLO 7)

MLO 7: Synthesize Chapter V by preparing the Summary of Findings, Conclusions, and Recommendations grounded on the interpreted results presented in Chapter IV and aligned with the research objectives.

Topic Learning Outcomes (TLO)

At the end of this module, the students should be able to:

TLO1: Explain the purpose, structure, and significance of Chapter V in completing a capstone research manuscript.

TLO 2: Prepare a concise Summary of Findings based on the statistical results presented in Chapter IV.

TLO 3: Formulate logical and evidence-based conclusions aligned with the research objectives and findings.

TLO 4: Develop practical and realistic recommendations that address the identified findings, limitations, and future improvements of the study.

TLO 5: Identify and avoid common errors in writing Chapter V.

Learning Objectives

Upon completion of this module, the students are expected to:

- explain the importance of Chapter V in research writing;



- summarize the major findings of the study accurately;
- formulate conclusions supported by research evidence;
- prepare recommendations that are realistic, actionable, and aligned with the study findings; and
- evaluate Chapter V for completeness, coherence, and academic quality.

Required Resources

Students should prepare the following before studying this module:

- Approved Chapters I-IV
- Revised Chapter IV Results and Discussion
- Statistical Analysis Outputs
- Research Objectives
- Laptop/Desktop Computer
- Word Processing Software
- Course Module
- Consultation Notes
- Adviser Recommendations

Module Overview

Chapter V serves as the culminating chapter of a capstone research project. It presents the overall findings of the study, draws conclusions based on analyzed data, and provides practical recommendations for stakeholders and future researchers. Unlike previous chapters, Chapter V does not introduce new information. Instead, it synthesizes the evidence gathered throughout the research process and demonstrates whether the objectives of the study have been successfully achieved.

A well-written Chapter V reflects the researchers' ability to analyze, synthesize, and interpret research outcomes critically and objectively. Every statement included in this chapter should be supported by the results presented in Chapter IV and remain consistent with the research objectives established in Chapter I.

Throughout this module, students will learn the principles, structure, and techniques involved in writing an effective Summary of Findings, Conclusions, and Recommendations while avoiding common writing errors frequently observed in capstone manuscripts.



Introduction

The completion of a capstone project does not end with developing a functional information system or presenting statistical results. Researchers are expected to communicate the significance of their findings by preparing a clear, organized, and evidence-based Chapter V.

Many beginning researchers mistakenly believe that Chapter V is simply a repetition of the Results and Discussion chapter. In reality, it requires higher-order thinking skills such as analysis, synthesis, evaluation, and decision-making. Researchers must determine what the findings mean, what conclusions may legitimately be drawn, and what recommendations should be proposed based on the evidence obtained during the study.

This module guides students in preparing a well-structured Chapter V that satisfies academic standards and institutional expectations while maintaining alignment with the objectives and scope of the study.

Lesson 1: Introduction to Chapter V

The completion of a capstone project does not end with the presentation of statistical analyses or the successful development of an information system. After collecting, analyzing, and interpreting the research data presented in Chapter IV, researchers are expected to synthesize the major findings of the study, formulate logical conclusions, and propose practical recommendations. These essential components are presented in **Chapter V**, which serves as the concluding chapter of the research manuscript.

Chapter V is regarded as one of the most important chapters because it demonstrates the researchers' ability to interpret the significance of their findings and communicate the contributions of the study to the intended beneficiaries, future researchers, and the academic community. A well-written Chapter V reflects critical thinking, analytical reasoning, and evidence-based decision-making.

Unlike previous chapters that focus on introducing the research problem, reviewing related literature, describing the methodology, or presenting statistical results, Chapter V integrates the entire research process into meaningful conclusions and actionable recommendations. Every statement included in this chapter must be directly supported by the evidence obtained during the study.

1.1 What is Chapter V?

Chapter V is the final chapter of a research manuscript that presents the **Summary of Findings, Conclusions, and Recommendations**. It synthesizes the most important results obtained from the study and explains what those results mean in relation to the research objectives.

Rather than presenting new data, Chapter V focuses on interpreting the overall outcome of the research. It answers the fundamental question:

"What can be concluded from the study, and what should be done based on the findings?"

For capstone projects, Chapter V also demonstrates whether the proposed information system successfully addressed the identified problem and achieved the objectives established in Chapter I.

1.2 Purpose of Chapter V

The primary purpose of Chapter V is to communicate the overall significance of the completed research. It allows researchers to synthesize the findings, draw logical conclusions, and recommend appropriate actions based on empirical evidence.

Specifically, Chapter V aims to:

- summarize the major findings of the study;
- determine whether the research objectives have been achieved;
- formulate conclusions supported by research evidence;
- provide practical recommendations for stakeholders;
- identify possible improvements for the developed system; and
- suggest directions for future studies.

Because Chapter V represents the culmination of the entire research process, it should present a coherent and evidence-based discussion rather than personal opinions or unsupported claims.

1.3 Components of Chapter V

Chapter V generally consists of three major sections.

A. Summary of Findings

This section presents a concise synthesis of the significant results obtained from the study. The findings should be organized according to the



research objectives or research questions and should be based solely on the results presented in Chapter IV.

The Summary of Findings should answer:

- What were the major findings of the study?
- What did the statistical results reveal?
- What important observations were obtained from the evaluation?

B. Conclusions

The Conclusions section presents the researchers' interpretation of the findings. Conclusions explain what the results mean and determine whether the objectives of the study have been achieved.

Unlike the Summary of Findings, conclusions require higher-level thinking because researchers must analyze and synthesize the evidence before making logical judgments.

Every conclusion should be supported by the findings of the study.

C. Recommendations

The Recommendations section provides practical suggestions based on the conclusions reached by the researchers.

Recommendations may be addressed to:

- the beneficiary organization;
- future system users;
- administrators;
- policy makers;
- future researchers; or
- software developers.

Recommendations should be realistic, feasible, and directly related to the findings of the study.

1.4 Relationship Between Chapters I–V

Each chapter of the research manuscript contributes to the successful completion of the study.

The relationship among the chapters may be summarized as follows:

Chapter	Primary Purpose
Chapter I	Identifies the research problem, objectives, and scope of the study.
Chapter II	Establishes the theoretical and conceptual foundation through related literature and studies.
Chapter III	Describes the research methodology and system development procedures.
Chapter IV	Presents and interprets the research findings and system evaluation results.
Chapter V	Synthesizes the findings, formulates conclusions, and provides recommendations.

Chapter V therefore serves as the culmination of all previous chapters by integrating the research evidence into meaningful outcomes.

1.5 Principles in Writing Chapter V

To produce a high-quality Chapter V, researchers should observe the following principles:

Evidence-Based Writing: Every statement must be supported by the findings presented in Chapter IV.

Objectivity: Researchers should avoid personal opinions, assumptions, or unsupported generalizations.

Logical Organization: The discussion should follow the sequence of the research objectives or research questions.

Consistency: The Summary of Findings, Conclusions, and Recommendations must be aligned with Chapters I through IV.

Clarity: Use concise, formal, and precise academic language.

1.6 Importance of Chapter V in Capstone Research

Chapter V demonstrates the practical value of the completed study.

For an Information Technology capstone project, this chapter allows researchers to determine whether the developed system:

- addressed the identified operational problems;
- satisfied the functional requirements;

- met software quality standards;
- achieved user acceptance;
- contributed to organizational improvement; and
- can be recommended for implementation.

Without a well-prepared Chapter V, readers may find it difficult to understand the overall contribution and significance of the research.

Lesson 2: Writing the Summary of Findings

The Summary of Findings is the first section of Chapter V and serves as a concise presentation of the most significant results obtained from the study. It provides readers with a clear overview of what the research discovered without repeating the detailed discussions and statistical analyses presented in Chapter IV.

A well-written Summary of Findings enables readers to understand the major outcomes of the study quickly and logically. It synthesizes the interpreted results according to the sequence of the research objectives or research questions, allowing the readers to determine whether the study has successfully addressed the identified problem.

Unlike Chapter IV, which explains and interprets statistical results in detail, the Summary of Findings focuses only on the essential information needed to support the conclusions and recommendations presented later in Chapter V.

2.1 What is the Summary of Findings?

The **Summary of Findings** is a concise synthesis of the major results obtained from the research. It highlights the most important findings without presenting detailed statistical computations, extensive interpretations, or new information.

Its primary purpose is to answer the following questions:

- What were the significant findings of the study?
- What did the collected data reveal?
- Were the research objectives achieved?
- What important observations were obtained from the respondents?

The Summary of Findings should present only factual statements that are directly supported by the results discussed in Chapter IV.



2.2 Purpose of the Summary of Findings

The Summary of Findings serves several important purposes in research writing.

Specifically, it aims to:

- present the major findings in a concise and organized manner;
- synthesize the results of the study according to the research objectives;
- provide the basis for formulating research conclusions;
- facilitate the understanding of the overall outcomes of the research; and
- establish the foundation for evidence-based recommendations.

Because it serves as the bridge between Chapter IV and the succeeding sections of Chapter V, the Summary of Findings should remain objective, accurate, and evidence-based.

2.3 Characteristics of an Effective Summary of Findings

An effective Summary of Findings possesses several important characteristics.

A. Objective: The discussion should present only the actual findings of the study without introducing personal opinions or assumptions.

B. Concise: Only the significant findings should be included. Avoid lengthy explanations and unnecessary details.

C. Organized: The findings should follow the same sequence as the research objectives or research questions presented in Chapter I.

D. Evidence-Based: Every statement should be supported by the results discussed in Chapter IV.

E. Consistent: The findings should remain consistent with the statistical analyses, tables, figures, and interpretations previously presented.

2.4 Organizing the Summary of Findings

The most effective way of organizing the Summary of Findings is according to the sequence of the research objectives.

A typical structure is shown below.



General Objective: Provide a brief statement describing whether the overall purpose of the study has been achieved.

Specific Objective 1: Summarize the major finding related to the first objective.

Specific Objective 2: Present the important findings corresponding to the second objective.

Specific Objective 3: Summarize the results associated with the third objective.

Continue until all research objectives have been addressed.

This organization helps readers easily determine how each objective was answered through the research.

2.5 Writing the Summary of Findings

The following procedure may be followed when preparing the Summary of Findings.

Step 1. Review the Research Objectives

Begin by reviewing the General Objective and Specific Objectives presented in Chapter I.

These objectives serve as the framework for organizing the Summary of Findings.

Step 2. Review Chapter IV

Read the Results and Discussion chapter carefully.

Identify the most significant findings related to each objective.

Avoid copying entire paragraphs from Chapter IV.

Step 3. Select Only the Major Findings

Choose only the most important results that directly answer the research objectives.

Minor observations and repetitive statements should not be included.

Step 4. Write Concise Statements

Summarize each finding using clear and concise academic language.

Avoid discussing statistical procedures or detailed computations.

Step 5. Verify Alignment

Ensure that every summarized finding corresponds to a research objective.
No finding should be included if it is unrelated to the objectives of the study.

2.6 Example of a Summary of Findings

Research Objective

To determine the software quality of the developed Aviation Maintenance Monitoring System using the ISO/IEC 25010 Product Quality Model.

Poor Example

The system obtained a weighted mean of 3.82 in Functional Suitability, 3.76 in Reliability, 3.88 in Security, 3.81 in Maintainability, and many other numerical results shown in Chapter IV.

Why is this poor?

- Merely repeats statistical values.
- Does not synthesize the findings.
- Reads like Chapter IV instead of Chapter V.

Good Example

The evaluation conducted by the IT experts indicated that the developed Aviation Maintenance Monitoring System demonstrated a high level of software quality across the evaluated ISO/IEC 25010 characteristics. The results suggest that the system provides appropriate functionality, reliable performance, effective security mechanisms, and maintainable software architecture suitable for its intended operational environment.

Why is this better?

- Synthesizes the findings.
- Focuses on the overall result.
- Uses evidence without repeating every numerical value.
- Leads naturally to the Conclusions section.



2.7 Common Mistakes in Writing the Summary of Findings

Beginning researchers often commit several mistakes when writing this section.

These include:

Repeating Chapter IV word-for-word

The Summary of Findings should synthesize information rather than copy previous discussions.

Including New Information

No new data, literature, interviews, or statistical analyses should be introduced.

Presenting Personal Opinions

Only research findings supported by evidence should be presented.

Ignoring the Research Objectives

Every finding should correspond to one or more research objectives.

Including Recommendations

Recommendations belong to a separate section and should not appear in the Summary of Findings.

Lesson 3: Formulating Research Conclusions

After summarizing the significant findings of the study, the next step in Chapter V is to formulate the research conclusions. The conclusions represent the researchers' final judgments regarding the outcomes of the study based on the evidence gathered and analyzed throughout the research process.

Unlike the Summary of Findings, which simply presents the significant results of the study, the Conclusions section goes beyond describing the findings. It explains what those findings mean, determines whether the research objectives have been achieved, and highlights the overall contribution of the study.

A well-formulated conclusion demonstrates the researchers' ability to synthesize information, interpret research evidence, and make logical decisions without introducing unsupported opinions or assumptions. Every conclusion must be grounded on the findings discussed in Chapter IV and should directly correspond to the research objectives established in Chapter I.

3.1 What are Research Conclusions?

Research conclusions are logical statements that explain the meaning and implications of the study's findings. They represent the researchers' interpretation of the results after carefully analyzing and synthesizing the evidence collected during the research.

The conclusions answer questions such as:

- What do the findings indicate?
- Were the research objectives achieved?
- What can be inferred from the results?
- What is the overall contribution of the study?

Research conclusions should not merely repeat the findings. Instead, they should present reasoned judgments that are directly supported by empirical evidence.

3.2 Purpose of Research Conclusions

The Conclusions section serves several important purposes in research writing.

Specifically, it aims to:

- determine whether the research objectives have been achieved;
- explain the significance of the research findings;
- summarize the overall outcome of the study;
- provide the basis for practical recommendations; and
- demonstrate the contribution of the research to the intended beneficiaries and future studies.

A carefully written conclusion allows readers to understand the value of the completed research without reviewing the entire manuscript.

3.3 Relationship Between Findings and Conclusions

One of the most common misconceptions among beginning researchers is that findings and conclusions are identical. Although they are closely related, they serve different purposes.

Summary of Findings	Research Conclusions
Presents the major results of the study.	Explains the meaning of those results.
Based on statistical findings and observations.	Based on logical interpretation of the findings.
Reports what was discovered.	Explains what can be concluded from the discoveries.
Does not interpret.	Interprets the findings objectively.

For example:

Finding

The respondents evaluated the developed system as highly acceptable in terms of Functional Suitability and Usability.

Conclusion

The developed system successfully met the functional requirements of the intended users and demonstrated its suitability for practical implementation within the beneficiary organization.

Notice that the conclusion does not simply repeat the finding. Instead, it explains its significance.

3.4 Principles in Formulating Research Conclusions

To produce meaningful conclusions, researchers should observe several important principles.

A. Base Every Conclusion on Evidence

Every conclusion must originate from the findings presented in Chapter IV.

Researchers should never formulate conclusions based on personal opinions or unsupported assumptions.



B. Align Conclusions with the Research Objectives

Each research objective should have a corresponding conclusion.

If the study contains five specific objectives, the conclusions should clearly indicate whether each objective has been achieved.

C. Avoid Introducing New Information

No new statistical data, references, interviews, or observations should be presented in this section.

Only evidence already discussed in the previous chapters should be used.

D. Write Clearly and Objectively

Research conclusions should be concise, direct, and supported by evidence.

Avoid emotional language, exaggeration, or unsupported claims.

E. Demonstrate Higher-Level Thinking

Conclusions should reflect analysis, synthesis, and evaluation rather than simple description.

Researchers should explain what the findings imply and why they are important.

3.5 Steps in Formulating Research Conclusions

Researchers may follow the procedure below when writing the Conclusions section.

Step 1. Review the Research Objectives

Read the General Objective and Specific Objectives presented in Chapter I.

These objectives provide the framework for organizing the conclusions.

Step 2. Review the Summary of Findings

Identify the significant findings corresponding to each objective.

Determine what these findings imply.

Step 3. Interpret the Findings

Analyze the results and determine their significance.

Ask questions such as:

- What do these findings indicate?
- What has been accomplished?
- What contribution has the study made?

Step 4. Write Logical Conclusions

Formulate concise statements supported by the research findings.

Avoid merely copying statements from the Summary of Findings.

Step 5. Verify Alignment

Ensure that every conclusion is directly supported by the findings and corresponds to the research objectives.

3.6 Example of a Research Conclusion

Research Objective

To evaluate the software quality of the developed Aviation Maintenance Monitoring System using the ISO/IEC 25010 Product Quality Model.

Poor Conclusion

The system received high ratings in Functional Suitability, Reliability, Security, and Maintainability.

Why is this poor?

- Simply repeats the findings.
- No interpretation is provided.
- Does not explain the significance of the results.

Better Conclusion

Based on the software quality evaluation, the developed Aviation Maintenance Monitoring System demonstrated that it satisfies the essential quality characteristics required for operational implementation. The consistently high ratings across the evaluated ISO/IEC 25010 characteristics indicate that the system is capable of supporting maintenance management activities effectively while maintaining reliability, security, and maintainability.

Why is this better?

- Interprets the findings.
- Demonstrates achievement of the research objective.
- Explains the significance of the evaluation.

3.7 Common Errors in Writing Research Conclusions

Researchers frequently commit the following mistakes:

Repeating the Summary of Findings

A conclusion should interpret the findings rather than restate them.

Presenting Personal Opinions

Statements such as:

"We believe..."

"In our opinion..."

have no place in research conclusions.

Making Unsupported Generalizations

Researchers should avoid claiming that the developed system is the "best" or "perfect" unless supported by evidence.



Introducing New Information

No additional statistical analyses, references, or observations should appear in this section.

Writing Conclusions That Are Unrelated to the Objectives

Every conclusion should correspond to one or more research objectives.

Lesson 4: Writing Evidence-Based Recommendations

The **Recommendations** section is the final component of Chapter V. It presents practical and feasible actions that may be undertaken based on the conclusions derived from the study. Recommendations are intended to provide guidance to the beneficiary organization, system users, administrators, policymakers, software developers, and future researchers regarding possible improvements, implementation strategies, and areas for further investigation.

A common misconception among beginning researchers is that recommendations are merely personal suggestions. In academic research, however, recommendations must be **evidence-based**. This means that every recommendation should originate from the findings and conclusions of the study rather than from assumptions, opinions, or preferences of the researchers.

Evidence-based recommendations demonstrate the researchers' ability to transform research findings into meaningful actions that contribute to solving real-world problems and improving existing systems or practices. Well-written recommendations increase the practical value of the research and strengthen its contribution to the academic and professional community.

4.1 What are Research Recommendations?

Research recommendations are **practical, logical, and actionable suggestions** proposed by the researchers after completing the study. These recommendations are formulated based on the conclusions drawn from the research findings and are intended to address identified needs, improve existing practices, or guide future developments.

The Recommendations section answers questions such as:

- What actions should be taken based on the findings?
- Who will benefit from these recommendations?



- How can the developed system or research be improved?
- What future studies should be conducted?

Unlike conclusions, which explain the meaning of the findings, recommendations focus on the actions that may be taken as a result of those findings.

4.2 Purpose of Research Recommendations

The Recommendations section serves several important purposes in a research manuscript. Specifically, it aims to:

- provide practical solutions based on the research findings;
- suggest improvements to the developed system, process, or organization;
- assist decision-makers in implementing appropriate actions;
- identify opportunities for future research; and
- maximize the practical impact and usefulness of the completed study.

A well-developed Recommendations section demonstrates that the researchers understand how the results of their study can be translated into meaningful actions.

4.3 Characteristics of Evidence-Based Recommendations

Effective recommendations possess several important characteristics.

A. Evidence-Based

Every recommendation should be supported by the findings and conclusions of the study.

Researchers should avoid making recommendations that have no basis in the research results.

B. Practical

Recommendations should be realistic and capable of being implemented using available resources.



Suggestions that are impossible to accomplish should be avoided.

C. Specific

Recommendations should clearly indicate the action to be taken and identify the intended beneficiaries.

Avoid vague statements such as:

"The system should be improved."

Instead, specify **what** should be improved and **why**.

D. Action-Oriented

Recommendations should encourage concrete actions rather than simply describing problems.

Action verbs such as **implement, develop, strengthen, enhance, establish, and conduct** are commonly used.

E. Relevant

Every recommendation should directly relate to the objectives, findings, and conclusions of the study.

Recommendations unrelated to the research should not be included.

4.4 Types of Research Recommendations

Depending on the nature of the study, recommendations may be directed toward different stakeholders.

A. Recommendations for the Beneficiary Organization

These recommendations focus on improving organizational operations through the implementation of the developed system or proposed solution.

Example:

The beneficiary organization may implement the developed Aviation Maintenance Monitoring System to improve maintenance scheduling, record management, and monitoring of aircraft maintenance activities.



B. Recommendations for System Administrators

These recommendations emphasize system maintenance, security, and operational management.

Example:

System administrators should regularly update user accounts, perform database backups, and monitor system performance to ensure continuous and secure operation.

C. Recommendations for End Users

These recommendations encourage proper utilization of the developed system.

Example:

End users should undergo orientation and training before using the system to maximize its features and minimize operational errors.

D. Recommendations for Software Developers

These recommendations identify possible technical enhancements based on the study findings.

Example:

Future versions of the system may incorporate mobile application support, real-time notification services, and cloud-based database synchronization to improve accessibility and operational efficiency.

E. Recommendations for Future Researchers

These recommendations encourage further investigation beyond the scope and limitations of the present study.

Example:

Future researchers may evaluate the long-term effectiveness of the developed system using a larger sample size or apply emerging technologies such as Artificial Intelligence, Machine Learning, or Predictive Analytics.



4.5 Steps in Writing Evidence-Based Recommendations

Researchers may follow the procedure below when preparing the Recommendations section.

Step 1. Review the Conclusions

Read the conclusions carefully to determine the issues, opportunities, and improvements identified during the study.

Step 2. Identify the Appropriate Stakeholders

Determine who will benefit from the recommendations.

Possible stakeholders include:

- Beneficiary organization
- Management
- Faculty
- Students
- Employees
- System administrators
- Software developers
- Future researchers

Step 3. Develop Actionable Recommendations

Write recommendations using clear action verbs.

Focus on feasible improvements rather than ideal situations.

Step 4. Verify Alignment

Ensure that every recommendation directly corresponds to one or more conclusions of the study.

Step 5. Review Practicality

Evaluate whether the recommendations are realistic, achievable, and appropriate for the intended beneficiaries.

4.6 Example of Evidence-Based Recommendations

Research Conclusion

The developed Aviation Maintenance Monitoring System demonstrated high acceptability in terms of Functional Suitability, Usability, Security, and Reliability.

Poor Recommendation

The system should become the best maintenance system in the Philippines.

Why is this poor?

- Unrealistic.
- Unsupported by the findings.
- Uses exaggerated language.

Better Recommendation

The beneficiary organization may adopt the developed Aviation Maintenance Monitoring System to enhance maintenance scheduling, improve record management, and support efficient monitoring of maintenance activities. Regular system maintenance and user training are likewise recommended to sustain system performance and maximize its operational benefits.

Why is this better?

- Supported by the conclusions.
- Practical and achievable.
- Clearly identifies the intended beneficiaries.
- Suggests specific actions.

4.7 Common Mistakes in Writing Recommendations

Researchers often commit the following mistakes:



Giving Recommendations Without Evidence

Recommendations should always be supported by the research findings and conclusions.

Being Too General

Statements such as:

"The organization should improve."

are vague and provide no clear direction.

Being Unrealistic

Recommendations requiring excessive financial resources, unavailable technology, or impractical conditions should be avoided unless justified.

Repeating the Conclusions

Recommendations should propose actions rather than restating research conclusions.

Ignoring the Scope of the Study

Recommendations should remain within the scope and limitations of the completed research.

Lesson 5: Common Errors in Chapter V

Chapter V is the culminating chapter of a research manuscript where the researchers synthesize the significant findings, formulate conclusions, and provide recommendations based on the evidence gathered throughout the study. Although Chapter V appears shorter than the previous chapters, it is often one of the most difficult sections to write because it requires higher-order thinking skills, including analysis, synthesis, evaluation, and decision-making.

Many beginning researchers commit errors that weaken the quality, credibility, and usefulness of their research. These mistakes commonly include repeating statistical

results instead of synthesizing findings, writing conclusions that are unsupported by evidence, providing unrealistic recommendations, and failing to align Chapter V with the research objectives.

Understanding these common errors enables researchers to produce a Chapter V that is coherent, evidence-based, and consistent with the entire research manuscript. By recognizing and avoiding these mistakes, students can effectively communicate the significance of their study and provide meaningful contributions to their chosen field.

5.1 Why Do Researchers Commit Errors in Chapter V?

Writing Chapter V requires researchers to integrate information gathered throughout the research process. Unlike earlier chapters that primarily describe, explain, or report information, Chapter V demands careful interpretation and sound judgment.

Researchers often commit errors because they:

- misunderstand the purpose of Chapter V;
- confuse findings with conclusions;
- provide recommendations without supporting evidence;
- fail to review the research objectives before writing; or
- rely on personal opinions instead of research findings.

Recognizing these common mistakes is the first step toward producing a well-written and academically sound Chapter V.

5.2 Common Error 1: Repeating Chapter IV Instead of Summarizing

One of the most frequent mistakes is copying paragraphs directly from the Results and Discussion chapter.

Poor Example

The system obtained a weighted mean of 3.87 in Functional Suitability, 3.81 in Reliability, 3.90 in Security, and 3.84 in Maintainability.

This statement merely repeats statistical results already presented in Chapter IV.

Better Example

The evaluation results indicated that the developed system demonstrated a high level of software quality across the assessed ISO/IEC 25010 characteristics, suggesting its suitability for operational implementation.

The improved version synthesizes the findings rather than repeating numerical values.

5.3 Common Error 2: Conclusions Not Supported by the Findings

Every conclusion must be based on evidence presented in Chapter IV.

Poor Example

The developed system is the most effective monitoring system available today.

This statement cannot be justified unless the study actually compared the system with all existing monitoring systems.

Better Example

The evaluation results indicate that the developed system effectively satisfied the identified functional requirements and achieved a high level of user acceptability within the context of the study.

This conclusion remains consistent with the actual scope and findings of the research.

5.4 Common Error 3: Recommendations Based on Personal Opinion

Recommendations should originate from the research conclusions rather than the researchers' preferences.

Poor Example

The organization should purchase expensive servers and the latest computers.

Unless the study specifically evaluated infrastructure needs, this recommendation lacks evidence.

Better Example

The beneficiary organization may consider upgrading its hardware infrastructure if future implementation requires increased system capacity and user demand.

This recommendation is practical and appropriately qualified.

5.5 Common Error 4: Misalignment with the Research Objectives

The Summary of Findings, Conclusions, and Recommendations should all correspond to the research objectives.

If a study has five specific objectives, Chapter V should clearly address all five.

Failure to discuss one or more objectives results in an incomplete Chapter V.

Researchers should review Chapter I before finalizing Chapter V to ensure complete alignment.

5.6 Common Error 5: Introducing New Information

Chapter V should not include:

- new statistical analyses;
- additional respondents;
- new literature or related studies;
- new survey results; or
- information that was never discussed in previous chapters.

The purpose of Chapter V is to synthesize and interpret existing findings—not to introduce new evidence.

5.7 Common Error 6: Using Subjective or Emotional Language

Academic writing requires objectivity.

Researchers should avoid expressions such as:

- "We believe..."



- "In our opinion..."
- "We are very happy that..."
- "The system is absolutely perfect."

Instead, use objective language.

Example

Instead of:

We believe the respondents loved the system.

Write:

The evaluation results indicate that the respondents expressed a high level of acceptance toward the developed system.

5.8 Common Error 7: Overgeneralizing the Results

Researchers should avoid making claims beyond the scope of the study.

Poor Example

This study proves that all schools in the Philippines should adopt this system.

Unless the research involved all schools nationwide, this conclusion is unsupported.

Better Example

The findings suggest that the developed system may be considered for implementation within institutions having operational requirements similar to those of the beneficiary organization.

The revised statement acknowledges the scope and limitations of the study.

5.9 Checklist Before Finalizing Chapter V

Before submitting the manuscript, researchers should verify the following:

Checklist Item	Yes	No
Are all research objectives addressed?	☐	☐
Are the findings summarized instead of copied?	☐	☐
Are all conclusions supported by Chapter IV?	☐	☐
Are recommendations based on the conclusions?	☐	☐
Is there any new information introduced in Chapter V?	☐	☐
Are academic writing standards consistently observed?	☐	☐
Are grammar, spelling, and formatting errors corrected?	☐	☐
Has the adviser reviewed the final Chapter V?	☐	☐

This checklist may be used as a self-evaluation tool before the manuscript is submitted for adviser review or final defense.

References

Institutional References

Philippine State College of Aeronautics. (2025). *IT411: Aviation Capstone Project 2 course syllabus* (Version 2, Revision 1, 1st Semester, S.Y. 2025–2026). Institute of Computer Studies.

Philippine State College of Aeronautics. (2013). *PhilSCA student research handbook: AY 2013–2016*.



Books

American Psychological Association. (2020). **Publication manual of the American Psychological Association** (7th ed.). American Psychological Association.
<https://doi.org/10.1037/0000165-000>

Creswell, J. W., & Creswell, J. D. (2023). **Research design: Qualitative, quantitative, and mixed methods approaches** (6th ed.). SAGE Publications.

Fraenkel, J. R., Wallen, N. E., & Hyun, H. H. (2019). **How to design and evaluate research in education** (10th ed.). McGraw-Hill Education.



Guided Learning Activity / Performance Task

Activity Title: Developing Chapter V: Summary of Findings, Conclusions, and Recommendations

Learning Outcomes Addressed

At the end of this activity, the students should be able to:

- synthesize the significant findings of the completed study;
- formulate evidence-based conclusions aligned with the research objectives;
- prepare practical recommendations supported by the conclusions; and
- evaluate the completeness and quality of Chapter V using a structured checklist.

Activity Description

Using your approved **Chapter IV (Results and Discussion)** and the **research objectives** presented in Chapter I, prepare a complete draft of **Chapter V** following the institutional format discussed in this module.

The draft must include:

- Summary of Findings
- Conclusions
- Recommendations

All sections should be evidence-based and properly aligned with the research objectives and findings.

Instructions:

Part I – Summary of Findings (30 points)

Prepare the Summary of Findings by following the sequence of your research objectives.

For each objective:

- identify the major finding;



- synthesize the result into one concise paragraph; and
- avoid copying discussions directly from Chapter IV.

Part II – Research Conclusions (30 points)

Formulate conclusions based on the summarized findings.

Each conclusion must:

- answer the corresponding research objective;
- interpret the significance of the findings;
- be supported by evidence; and
- avoid introducing new information.

Part III – Evidence-Based Recommendations (20 points)

Develop practical recommendations for the following stakeholders whenever applicable:

- Beneficiary Organization
- System Administrator
- End Users
- Future Researchers

Recommendations should:

- originate from the conclusions;
- be realistic;
- be actionable; and
- remain within the scope of the study.

Part IV – Chapter V Self-Evaluation Checklist (20 points)

Before submission, evaluate your Chapter V using the checklist below.

Evaluation Item	Yes	No
The Summary of Findings follows the research objectives.	☐	☐
The findings are synthesized rather than copied.	☐	☐
Every conclusion is supported by the findings.	☐	☐

Recommendations are based on the conclusions.	☐	☐
No new information is introduced.	☐	☐
Academic language is consistently observed.	☐	☐
Grammar and formatting have been reviewed.	☐	☐
Chapter V is complete and logically organized.	☐	☐

Expected Output

Each research group shall submit:

- ☒ Complete Chapter V
 - Summary of Findings
 - Conclusions
 - Recommendations

along with the accomplished Chapter V Self-Evaluation Checklist.

Assessment Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Improvement (1)	Points
Summary of Findings	Findings are accurately synthesized, well-organized, and fully aligned with the research objectives.	Minor revisions needed in organization or clarity.	Several findings lack synthesis or alignment.	Findings are incomplete or copied directly from Chapter IV.	30
Research Conclusions	Conclusions are evidence-based, logical, and directly supported by the findings.	Conclusions are generally supported but require refinement.	Some conclusions are weak or insufficiently supported.	Conclusions are unsupported, vague, or inconsistent with the findings.	30
Recommendations	Recommendations are practical, realistic, evidence-based, and appropriate.	Recommendations are relevant but need greater specificity or practicality.	Recommendations are only partially related to the conclusions.	Recommendations are unrealistic, unsupported, or unrelated to the study.	20

	for the intended stakeholders.				
Organization and Academic Writing	Chapter V is coherent, logically organized, grammatically correct, and follows the required format.	Minor errors in grammar, organization, or formatting.	Several writing or formatting errors affect readability.	Numerous errors make the chapter difficult to understand.	20
TOTAL					100 Points



MODULE 2

Research Day: Consultation and Revision Workshop

Week: 8 8 | Time Allotment: 3 Hours

Course Learning Outcomes (CLO)

CLO 6: Produce complete capstone documentation and technical deliverables using the prescribed NAAP/ICS capstone layout, citation requirements, formatting standards, appendices, revision logs, source files, and system files with institutional compliance.

CLO 8: Demonstrate professionalism, teamwork, ethical research practice, proper AI-tool disclosure, data privacy awareness, and respect for intellectual property throughout system development, defense, deployment, and clearance processes.

Module Learning Outcome (MLO 8)

Revise the manuscript and system based on adviser, statistician, and panel feedback while updating the revision matrix and consultation records.

Topic Learning Outcomes (TLO)

TLO 1: Apply adviser, statistician, and/or panel feedback to improve documentation and system outputs.

TLO 2: Update the revision matrix, consultation log, and manuscript sections based on verified feedback.

Learning Objectives

At the end of this module, the students should be able to:

1. document adviser recommendations accurately during consultation;
2. revise the manuscript and system based on verified feedback;
3. update the revision matrix and consultation records systematically;
4. demonstrate professionalism during research consultations; and
5. prepare improved capstone outputs for the succeeding module.



Required Resources

1. Latest Capstone Manuscript
2. Revision Matrix
3. Consultation Log Sheet
4. Adviser Recommendation Notes
5. Laptop/Desktop Computer
6. Learning Management System (LMS)
7. Google Drive / Google Classroom
8. Prototype/System (if applicable)

Module Overview

This module is conducted as a **Research Day Consultation Workshop**, where student groups work closely with their assigned advisers to review the progress of their capstone projects. Rather than introducing new theoretical concepts, this module emphasizes manuscript refinement, system improvement, consultation documentation, and evidence-based revision tracking.

Students are expected to document every recommendation received from their adviser, statistician, and, when applicable, panel members. These recommendations should then be reflected in the revision matrix and incorporated into the revised manuscript and system prototype. Through continuous consultation and revision, students strengthen the quality, completeness, and institutional compliance of their capstone project before proceeding to the succeeding stages of development and defense.

Introduction

Revision is a continuous and essential process in capstone research. Every consultation provides opportunities to improve the technical quality, organization, and academic integrity of both the research manuscript and the developed system.

During this Research Day session, students are expected to engage actively in adviser consultation, document all recommendations, revise identified sections of the manuscript, and maintain complete consultation records. By systematically tracking revisions, students ensure that all recommendations are properly addressed and that the capstone project progresses according to institutional standards.



Learning Content

Lesson 1. Consultation Documentation

Consultation is an essential part of the capstone research process. It provides students with the opportunity to receive guidance, recommendations, and technical advice from their assigned adviser regarding the research manuscript, system development, and other project requirements. Every consultation serves as a basis for improving the quality and completeness of the capstone project.

To ensure that all recommendations are properly implemented, students should maintain accurate consultation records. These records should include the consultation date, topics discussed, recommendations provided by the adviser, agreed actions, and the status of implementation. Proper documentation helps monitor project progress, prevents important comments from being overlooked, and provides evidence that revisions were completed based on adviser feedback.

Students are encouraged to review their consultation records regularly and update their revision matrix after every consultation. Maintaining complete and organized documentation promotes accountability, facilitates communication between the research group and the adviser, and supports the successful completion of the capstone project.

Best Practice

Record adviser recommendations immediately after each consultation. Do not rely on memory alone. Keeping complete consultation records makes it easier to track revisions and monitor the overall progress of the project.

Lesson 2. Adviser Feedback

Adviser feedback plays a vital role in improving the quality of both the research manuscript and the developed system. During each consultation, students receive comments, suggestions, and recommendations that help ensure the study complies with institutional standards and achieves its research objectives. Feedback may also be provided by the statistician and panel members, particularly on research methodology, data analysis, system functionality, documentation, and technical presentation.

Students should carefully review every recommendation and determine the appropriate revisions needed. Feedback should not be viewed as criticism but as an opportunity to strengthen the research and enhance the overall quality of



the capstone project. Each recommendation should be documented, evaluated, and addressed through appropriate revisions to the manuscript, system prototype, or supporting documents.

By analyzing adviser, statistician, and panel feedback objectively, students can make informed decisions, resolve identified issues, and continuously improve their capstone project throughout the research process.

Best Practice

Read all recommendations carefully before making revisions. If a comment is unclear, seek clarification from your adviser during the consultation. Always implement revisions systematically and record the completed actions in the Revision Matrix for future reference.

Lesson 3. Revision Matrix Update

The **Revision Matrix** is an important document used to monitor the progress of manuscript and system revisions throughout the capstone process. It serves as an organized record of all recommendations provided by the adviser, statistician, or panel members, together with the corresponding actions taken by the research group. Maintaining an updated Revision Matrix demonstrates that the group has systematically addressed all comments and continuously improved the quality of the project.

A complete Revision Matrix should clearly indicate the recommendation received, the specific revision made, supporting evidence such as the revised page or section, and the current status of implementation. Updating the matrix after every consultation helps researchers track completed tasks, identify pending revisions, and prepare accurate documentation for future consultations and project evaluations.

Students are encouraged to review the Revision Matrix regularly to ensure that all recommendations have been addressed and properly documented before submitting the revised manuscript and system for further evaluation.

Best Practice

Update the Revision Matrix immediately after every consultation while the recommendations are still fresh and accurate. Always indicate the exact manuscript page, section, or system component where the revision was made to make verification easier for the adviser and panel members.

Recommendation	Action Taken	Supporting Evidence	Status	Verified By
Revise the Statement of the Problem.	Updated Research Question No. 3.	Chapter I, p. 5	Completed	Adviser

Lesson 4. Manuscript Revision

Manuscript revision is a continuous process that ensures the capstone research meets academic standards and institutional requirements. After every consultation, students should carefully review the comments and recommendations provided by the adviser, statistician, and panel members, then incorporate the necessary revisions into the appropriate sections of the manuscript.

Revisions may involve updating **Chapters I-V**, correcting references and citations, improving tables and figures, revising appendices, or refining technical documentation. Each revision should be based on verified consultation feedback and should remain consistent with the approved research objectives, methodology, and system implementation. Students should also ensure that all changes are reflected in the latest version of the manuscript before submission.

Regular manuscript revision improves the clarity, accuracy, and overall quality of the research while preparing the project for the succeeding consultation, defense, and final submission.

Best Practice

Revise only the sections identified during consultation and verify that every correction has been incorporated into the latest manuscript version. Before submitting the revised manuscript, proofread the document for grammar, formatting, consistency, and completeness to ensure that it complies with the prescribed NAAP/ICS capstone format.

Revision Checklist	✓
Chapters I–V updated based on adviser comments	☐
References and citations verified	☐
Tables and figures reviewed	☐
Appendices updated	☐
Revision Matrix updated	☐
Latest manuscript version saved	☐

Lesson 5. Consultation Best Practices

Consultation is an essential component of Capstone Project 2 because it provides students with expert guidance in improving both the research manuscript and the developed system. To maximize every consultation session, students should come prepared by reviewing their current outputs, organizing their questions, and bringing the necessary documents, such as the latest manuscript, revision matrix, consultation log, and system prototype.

During the consultation, students are expected to practice active listening, take accurate notes, ask clarifying questions when necessary, and respond professionally to the recommendations given by the adviser. Every comment should be documented and reflected in the corresponding revisions. Prompt implementation of verified recommendations demonstrates commitment to continuous improvement and helps ensure that the project progresses according to the approved schedule.

Professional consultation practices strengthen communication between the research group and the adviser, promote accountability, and contribute to the successful completion of the capstone project.

Best Practice

Prepare all required documents before every consultation, arrive on time, listen carefully to every recommendation, and record important comments accurately. After the consultation, implement the agreed revisions promptly and



update the Revision Matrix and Consultation Log to maintain complete and organized project documentation.

Consultation Checklist

- ☐ Latest manuscript is updated and printed (if required).
- ☐ Revision Matrix is updated.
- ☐ Consultation Log is ready for recording adviser comments.
- ☐ System prototype is available for demonstration (if applicable).
- ☐ Questions and concerns have been prepared.
- ☐ Required supporting documents are complete.
- ☐ Agreed revisions are implemented after the consultation.

References:

Institutional References

- Philippine State College of Aeronautics. (2025). *IT411: Aviation Capstone Project 2 course syllabus* (Version 2, Revision 1, 1st semester, S.Y. 2025–2026). Institute of Computer Studies.
- Philippine State College of Aeronautics. (2013). *PhilSCA student research handbook: AY 2013–2016*.

Style Guide

- American Psychological Association. (2020). **Publication manual of the American Psychological Association** (7th ed.). American Psychological Association. <https://doi.org/10.1037/0000165-000> (American Psychological Association)

Research Methodology and Academic Writing

- Creswell, J. W., & Creswell, J. D. (2023). **Research design: Qualitative, quantitative, and mixed methods approaches** (6th ed.). SAGE Publications.



Fraenkel, J. R., Wallen, N. E., & Hyun, H. H. (2019). **How to design and evaluate research in education** (10th ed.). McGraw-Hill Education.

Technical Writing and Research Documentation

Day, R. A., & Gastel, B. (2024). **How to write and publish a scientific paper** (10th ed.). Cambridge University Press.

Research Ethics

Resnik, D. B. (2020). **What is ethics in research & why is it important?** National Institute of Environmental Health Sciences.
https://www.niehs.nih.gov/research/resources/bioethics/whatis?utm_source

Academic Integrity

International Center for Academic Integrity. (2021). **The fundamental values of academic integrity** (3rd ed.).
https://academicintegrity.org/aws/ICAI/pt/sp/home_page



Guided Learning Activity

Activity Title: Research Day Consultation and Revision Workshop

Directions: Each capstone group shall accomplish the following tasks during the scheduled Research Day:

Part I: Present the latest version of the manuscript and system prototype to the assigned adviser.

Part II: Record all adviser recommendations using the official Consultation Log.

Part III: Update the Revision Matrix by indicating:

- Recommendation
- Action Taken
- Supporting Evidence
- Revision Status

Part IV: Revise the manuscript and/or system based on the consultation.

Part V: Submit the updated documents through the Learning Management System.

Expected Outputs

Each group shall submit the following:

- Updated Capstone Manuscript
- Updated Revision Matrix
- Consultation Log
- Revised Prototype/System (if applicable)
- Other supporting documents required by the adviser



Formative Assessment

Research Day Progress Monitoring

The instructor shall evaluate the following:

- Consultation Attendance
- Consultation Documentation
- Updated Revision Matrix
- Progress of Manuscript Revision
- Compliance with Adviser Recommendations

Summative Assessment: Research Day Consultation Portfolio

Students shall submit a portfolio containing:

- Consultation Log
- Updated Revision Matrix
- Revised Manuscript
- Supporting Evidence of Revisions

This portfolio demonstrates the group's progress in implementing adviser recommendations and maintaining complete documentation throughout the capstone process.

Rubric

Criteria	Excellent (4)	Good (3)	Fair (2)	Needs Improvement (1)	Weight
Consultation Documentation	Complete, accurate, and properly organized consultation records	Minor omissions	Several missing details	Incomplete documentation	20%
Revision Matrix	All recommendations documented with corresponding actions and evidence	Minor revisions needed	Partially completed	Lacks organization or completeness	25%
Quality of Manuscript Revision	Revisions fully address adviser feedback and improve the manuscript	Most revisions completed	Some revisions incomplete	Minimal or no revisions made	35%
Professionalism and Compliance	Demonstrates professionalism, teamwork, punctuality, and complete compliance	Minor issues observed	Inconsistent participation	Poor compliance and professionalism	20%
TOTAL					100%



MODULE 3

Preparing for the Mock Final Defense

Week: 9 Time Allotment: 3 Hours

Course Learning Outcomes (CLO)

CLO 2: Develop, refine, and demonstrate IT-based capstone systems using approved system designs, development tools, and stakeholder requirements to address target beneficiary needs with at least **80% functionality** based on the system evaluation rubric.

CLO 6: Produce complete capstone documentation and technical deliverables using the prescribed NAAP/ICS capstone layout, citation requirements, formatting standards, appendices, revision logs, source files, and system files with institutional compliance.

CLO 7: Present and defend the capstone system and manuscript before advisers, panel members, users, and/or institutional evaluators using oral presentation, system demonstration, visual materials, and evidence-based responses with confidence, technical accuracy, and professionalism.

CLO 8: Demonstrate professionalism, teamwork, ethical research practice, proper AI-tool disclosure, data privacy awareness, and respect for intellectual property throughout system development, defense, deployment, and clearance processes.

Module Learning Outcome (MLO 9)

Prepare the capstone group for the mock final defense by evaluating the system, manuscript, presentation flow, and evidence of compliance.

Topic Learning Outcomes (TLO)

TLO 1: Demonstrate the working system and explain how its functions address the research objectives and user requirements.

TLO 2: Critique the manuscript, presentation slides, and system demonstration using defense rubrics and peer-review feedback.



Learning Objectives

At the end of this module, the students should be able to:

1. prepare the complete manuscript, presentation slides, and supporting documents for the mock final defense;
2. demonstrate the functionality of the developed system confidently and effectively;
3. present the capstone project using organized and professional presentation techniques;
4. respond appropriately to simulated panel questions using evidence from the study; and
5. evaluate and improve the presentation through constructive peer-review feedback.

Preparing for the Mock Final Defense

Lesson 1: Mock Defense Overview

The Mock Final Defense is a practice presentation conducted before the actual final defense. It allows the research group to evaluate the completeness of the manuscript, assess the functionality of the developed system, and improve presentation skills before appearing before the official panel. Through this activity, students can identify areas for improvement and gain confidence in presenting their capstone project.

Best Practice

Treat the mock defense as the actual final defense by preparing all required documents, practicing the presentation, and responding professionally to feedback.

Lesson 2: Preparing Effective Presentation Slides

Presentation slides should summarize the essential parts of the capstone project without copying the entire manuscript. Slides should be clear, organized, and visually appealing, highlighting the research problem, objectives, methodology, system features, results, conclusions, and recommendations. Simple layouts with readable text and relevant visuals help communicate information more effectively.



Best Practice

Use concise bullet points, readable fonts, and screenshots or diagrams instead of lengthy paragraphs. Allow the presenter to explain the details rather than reading directly from the slides.

Lesson 3: Demonstrating the Working System

The system demonstration allows the panel to evaluate whether the developed system satisfies the approved requirements and research objectives. Students should demonstrate the major system features, explain the workflow, and present how the system addresses the identified problem. The demonstration should be organized, functional, and free from unnecessary interruptions.

Best Practice

Test the system thoroughly before the presentation and prepare backup files, databases, and demonstration data to avoid technical problems during the mock defense.

Lesson 4: Responding to Panel Questions

Panel questions are intended to assess the students' understanding of the research and the developed system. Students should listen carefully, answer honestly and confidently, and support their responses using information from the manuscript and system implementation. If clarification is needed, students should respectfully ask the panel to restate the question.

Best Practice

Answer only what is being asked, remain professional, and support every response with evidence from the study rather than personal opinions.

Lesson 5: Peer Evaluation

Peer evaluation provides constructive feedback that helps improve the quality of the manuscript, presentation, and system demonstration before the actual final defense. By evaluating one another using the prescribed rubric, students can identify strengths, recognize weaknesses, and recommend practical improvements that enhance overall presentation performance.



Best Practice

Provide objective and respectful feedback based on the evaluation criteria. Focus on helping fellow researchers improve rather than criticizing their work.

References:

Institutional References

Philippine State College of Aeronautics. (2025). *IT411: Aviation Capstone Project 2 course syllabus* (Version 2, Revision 1, 1st semester, S.Y. 2025–2026). Institute of Computer Studies.

Philippine State College of Aeronautics. (2013). *PhilSCA student research handbook: AY 2013–2016*.

Style Guide

American Psychological Association. (2020). **Publication manual of the American Psychological Association** (7th ed.). American Psychological Association. <https://doi.org/10.1037/0000165-000> (American Psychological Association)

Research Methodology and Academic Writing

Creswell, J. W., & Creswell, J. D. (2023). **Research design: Qualitative, quantitative, and mixed methods approaches** (6th ed.). SAGE Publications.

Fraenkel, J. R., Wallen, N. E., & Hyun, H. H. (2019). **How to design and evaluate research in education** (10th ed.). McGraw-Hill Education.

Technical Writing and Research Documentation

Day, R. A., & Gastel, B. (2024). **How to write and publish a scientific paper** (10th ed.). Cambridge University Press.

Research Ethics

Resnik, D. B. (2020). **What is ethics in research & why is it important?** National Institute of Environmental Health Sciences. https://www.niehs.nih.gov/research/resources/bioethics/whatis?utm_source



Academic Integrity

International Center for Academic Integrity. (2021). **The fundamental values of academic integrity** (3rd ed.).
https://academicintegrity.org/aws/ICAI/pt/sp/home_page

Guided Learning Activity

Mock Final Defense Simulation

Each capstone group shall:

- present the completed manuscript;
- demonstrate the working system;
- answer simulated panel questions;
- receive peer evaluation; and
- revise the presentation based on the feedback received.

Formative Assessment

- Peer Evaluation Form
- Mock Defense Observation Notes

Summative Assessment

Mock Final Defense Presentation

Rubric E. Mock-Up Final Defense

Criteria	Excellent (4)	Very Satisfactory (3)	Satisfactory (2)	Needs Improvement (1)
Preparedness and flow	Presentation is complete, organized, timed, and well coordinated.	Generally organized with minor issues.	Partially organized but needs improvement.	Unprepared and poorly coordinated.
System demonstration readiness	System is functional and clearly demonstrates required features.	System is functional with minor issues.	System has partial functionality.	System is incomplete or not demonstrable.

Criteria	Excellent (4)	Very Satisfactory (3)	Satisfactory (2)	Needs Improvement (1)
Research and technical explanation	Methodology, results, design, and technical decisions are explained accurately.	Mostly accurate with minor gaps.	Some explanations are incomplete.	Explanations are inaccurate or missing.
Peer/panel feedback use	Feedback is actively gathered and converted into concrete improvement actions.	Most feedback is noted and used.	Feedback is partially considered.	Feedback is ignored or undocumented.
Professionalism	Presenters show confidence, teamwork, and proper defense decorum.	Generally professional.	Some lapses in professionalism.	Unprofessional or uncoordinated.



MODULE 4

Final Defense Presentation and Technical Justification

Week: 10 | Time Allotment: 3 Hours

Course Learning Outcomes (CLO)

CLO 2: Develop, refine, and demonstrate IT-based capstone systems using approved system designs, development tools, and stakeholder requirements to address target beneficiary needs with at least **80% functionality** based on the system evaluation rubric.

CLO 6: Produce complete capstone documentation and technical deliverables using the prescribed NAAP/ICS capstone layout, citation requirements, formatting standards, appendices, revision logs, source files, and system files with institutional compliance.

CLO 7: Present and defend the capstone system and manuscript before advisers, panel members, users, and/or institutional evaluators using oral presentation, system demonstration, visual materials, and evidence-based responses with confidence, technical accuracy, and professionalism.

CLO 8: Demonstrate professionalism, teamwork, ethical research practice, proper AI-tool disclosure, data privacy awareness, and respect for intellectual property throughout system development, defense, deployment, and clearance processes.

Module Learning Outcome (MLO 10)

Defend the completed capstone system and manuscript by presenting the methodology, findings, system functionality, and technical decisions while responding professionally to panel questions.

Topic Learning Outcomes (TLO)

TLO 1: Defend the research methodology, results, system design, system functionality, and technical decisions before the panel.

TLO 2: Respond accurately and professionally to panel questions, comments, and revision instructions.



Learning Objectives

At the end of this module, the students should be able to:

1. present the completed capstone manuscript following the prescribed defense format;
2. justify the research methodology, system design, and technical decisions using research evidence;
3. demonstrate the functionality of the developed system confidently and professionally;
4. respond accurately and objectively to panel questions and recommendations; and
5. exhibit professionalism, teamwork, and confidence throughout the final defense presentation.

Required Resources

- Final Capstone Manuscript
- Presentation Slides
- Working System Prototype
- Laptop/Desktop Computer
- Multimedia Projector
- Internet Connection (if applicable)
- Backup Copy of the System
- Learning Management System (LMS)

Module Overview

This module prepares students for the **Final Defense Presentation**, where they formally present and defend their completed capstone project before the evaluation panel. Students are expected to demonstrate the functionality of the developed system, justify the research methodology and technical decisions, and respond professionally to questions and recommendations from the panel members. Successful completion of the final defense signifies that the project has met the academic, technical, and institutional requirements of the Capstone Project 2 course.



Introduction

The Final Defense is the culminating activity of the capstone course. It allows students to demonstrate not only the quality of their research manuscript and developed system but also their understanding of the entire research and development process. Through a professional oral presentation and system demonstration, students validate the significance, functionality, and contribution of their project while receiving constructive feedback from the evaluation panel for final improvement.

Lesson 1: Structure of the Final Defense

The Final Defense follows the prescribed institutional format and includes the presentation of the research background, objectives, methodology, system demonstration, findings, conclusions, and recommendations. Students should present the project in a logical sequence while observing the allotted presentation time.

Best Practice

Follow the prescribed presentation outline and ensure that all required sections are discussed clearly and within the allotted time.

Lesson 2: Defending the Research Methodology

Students should justify the research design, data-gathering procedures, statistical treatment, and system development methodology used in the study. Every explanation should be consistent with the approved manuscript and supported by research evidence.

Best Practice

Explain the reasons for selecting the methodology rather than simply describing its procedures.

Lesson 3: Defending the Developed System

The system demonstration should highlight the major features, workflow, and functionality of the developed application. Students should explain how the



system satisfies the identified user requirements and research objectives while demonstrating stable system performance.

Best Practice

Demonstrate only the essential features related to the research objectives and prepare backup files before the presentation.

Lesson 4: Handling Panel Questions

Panel questions are intended to evaluate the students' understanding of the research and system implementation. Students should answer confidently, honestly, and respectfully while supporting their responses with evidence from the manuscript and system outputs.

Best Practice

Listen carefully, answer directly, and ask for clarification if a question is not fully understood.

Lesson 5: Professional Presentation Skills

Professionalism during the Final Defense reflects the students' preparedness and confidence. Proper attire, effective communication, teamwork, and respect for the evaluation panel contribute to a successful defense.

Best Practice

Maintain eye contact, speak clearly, observe proper etiquette, and acknowledge panel recommendations professionally.

References

Institutional References

Philippine State College of Aeronautics. (2025). *IT411: Aviation Capstone Project 2 course syllabus* (Version 2, Revision 1, 1st semester, S.Y. 2025–2026). Institute of Computer Studies.

Philippine State College of Aeronautics. (2013). *PhilSCA student research handbook: AY 2013–2016*.



Style Guide

American Psychological Association. (2020). **Publication manual of the American Psychological Association** (7th ed.). American Psychological Association. <https://doi.org/10.1037/0000165-000> (American Psychological Association)



Guided Learning Activity

Final Defense Simulation

Each research group shall:

- present the completed capstone manuscript;
- demonstrate the developed system;
- justify the research methodology and technical decisions;
- respond to simulated panel questions; and
- receive feedback from the instructor and classmates for improvement before the official defense.

Formative Assessment

- Defense Presentation Checklist
- Peer Observation Notes
- Instructor Feedback

Summative Assessment

Final Defense Presentation and System Demonstration

Each group shall present the completed capstone project before the evaluation panel and demonstrate the functionality of the developed system. Students shall be evaluated based on the quality of the presentation, technical justification, system demonstration, and responses to panel questions.

Rubrics

Rubric F. Final Defense - System and Manuscript

Criteria	Excellent (4)	Very Satisfactory (3)	Satisfactory (2)	Needs Improvement (1)
System functionality and quality	System is functional, stable, and aligned with objectives and user needs.	Functional with minor technical issues.	Partially functional with notable issues.	Incomplete or nonfunctional system.
Manuscript completeness and compliance	Manuscript follows the NAAP/ICS layout and contains complete chapters, references, appendices, and evidence.	Mostly complete with minor revisions needed.	Several sections or formatting details need revision.	Major sections are missing or noncompliant.
Research validity and results	Methodology, instruments, data, and results are valid, reliable, and well defended.	Generally valid with minor issues.	Partially valid or weakly defended.	Invalid, unsupported, or poorly defended.

Criteria	Excellent (4)	Very Satisfactory (3)	Satisfactory (2)	Needs Improvement (1)
Response to panel questions	Responses are accurate, concise, evidence-based, and professional.	Responses are generally accurate.	Responses are partially accurate or incomplete.	Responses are unclear or unsupported.
Professional oral defense	Presentation is confident, organized, time-conscious, and technically accurate.	Mostly organized and professional.	Understandable but needs better coordination.	Unprepared or unprofessional.



MODULE 5

Post-Defense Revision and Compliance Preparation

Week: 11 | Time Allotment: 3 Hours

Course Learning Outcomes (CLO)

CLO 2: Develop, refine, and demonstrate IT-based capstone systems using approved system designs, development tools, and stakeholder requirements to address target beneficiary needs with at least 80% functionality based on the system evaluation rubric.

CLO 6: Produce complete capstone documentation and technical deliverables using the prescribed NAAP/ICS capstone layout, citation requirements, formatting standards, appendices, revision logs, source files, and system files with institutional compliance.

CLO 8: Demonstrate professionalism, teamwork, ethical research practice, proper AI-tool disclosure, data privacy awareness, and respect for intellectual property throughout system development, defense, deployment, and clearance processes.

Module Learning Outcome (MLO 11)

Revise the manuscript and system after the final defense or re-defense by prioritizing panel comments and completing all compliance requirements.

Topic Learning Outcomes (TLO)

TLO 1: Apply panel recommendations to improve the capstone manuscript, system, and supporting documents.

TLO 2: Complete all institutional compliance requirements, revision records, and routing documents before final submission.

Learning Objectives

At the end of this module, the students should be able to:

1. analyze and prioritize panel comments received during the final defense;
2. revise the manuscript and system based on verified recommendations;
3. update appendices, supporting documents, and revision records;



4. complete institutional compliance and routing requirements; and
5. prepare the capstone project for final submission.

Required Resources

- Final Defense Evaluation Sheet
- Panel Recommendation Sheet
- Revised Capstone Manuscript
- Revision Matrix
- Consultation Log
- Routing Slip
- Laptop/Desktop Computer
- Learning Management System (LMS)

Module Overview

This module focuses on the revision and completion phase following the Final Defense. Students are expected to analyze panel recommendations, revise the manuscript and system accordingly, update supporting documents, and complete all institutional compliance requirements. The module emphasizes proper documentation, revision tracking, and preparation for routing and final submission.

Introduction

The Final Defense is not the end of the capstone process. Most research projects require revisions based on the recommendations of the evaluation panel. These revisions strengthen the technical quality, academic content, and institutional compliance of the project. Through systematic revision and proper documentation, students ensure that the final manuscript and developed system meet the standards required for approval and archiving.

Lesson 1: Understanding Final Defense Comments

Final defense comments provide guidance for improving the manuscript and the developed system. Students should review every recommendation carefully and identify the sections requiring revision. Understanding the intent of each comment helps ensure that revisions are accurate and complete.



Best Practice

Group similar comments together before starting revisions to avoid duplicate work and ensure consistency.

Lesson 2: Prioritizing Revisions

Not all revisions have the same level of urgency. Critical revisions affecting the research objectives, methodology, findings, or system functionality should be completed first, followed by formatting and editorial corrections.

Best Practice

Prepare a revision plan that identifies high-priority, medium-priority, and minor revisions.

Lesson 3: Updating Appendices

After revising the manuscript, students should also update the appendices and supporting documents, including survey instruments, consultation records, validation forms, screenshots, source codes (if required), and other project evidence.

Best Practice

Ensure that every appendix referenced in the manuscript is complete, properly labeled, and reflects the latest version of the project.

Lesson 4: Compliance Requirements

Institutional compliance includes completing all required forms, obtaining necessary signatures, updating revision records, and preparing the documents required by the department before final submission.

Best Practice

Use a compliance checklist to monitor completed and pending requirements before submission.



Lesson 5: Preparing for Routing

Routing is the formal process of securing approvals from the required offices before the manuscript is accepted for archiving. Students should verify that all documents are complete before submitting them for routing.

Best Practice

Arrange all documents according to the prescribed institutional sequence to avoid delays during the routing process.

References

Institutional References

Philippine State College of Aeronautics. (2025). *IT411: Aviation Capstone Project 2 course syllabus* (Version 2, Revision 1, 1st semester, S.Y. 2025–2026). Institute of Computer Studies.

Philippine State College of Aeronautics. (2013). *PhilSCA student research handbook: AY 2013–2016*.

Style Guide

American Psychological Association. (2020). **Publication manual of the American Psychological Association** (7th ed.). American Psychological Association. <https://doi.org/10.1037/0000165-000> (American Psychological Association)



Guided Learning Activity

Activity Title: Post-Defense Revision Workshop

Directions: Each research group shall:

- review the Final Defense Evaluation Sheet;
- classify panel comments according to priority;
- update the Revision Matrix;
- revise the manuscript and system;
- update appendices and supporting documents; and
- prepare the complete compliance folder for routing.

Expected Outputs

Each group shall submit:

- Revised Capstone Manuscript
- Updated Revision Matrix
- Updated Appendices
- Compliance Checklist
- Routing Documents

Formative Assessment

- Panel Comment Analysis
- Updated Revision Matrix
- Compliance Checklist

Summative Assessment

Post-Defense Revision Portfolio

Each group shall submit a **Post-Defense Revision Portfolio** containing all revised documents, supporting evidence, and compliance requirements for adviser verification.

Rubrics

Rubric G – Post-Defense Revision and Compliance Portfolio

Rubric G. Re-defense or Revised System and Documentation Presentation

Criteria	Excellent (4)	Very Satisfactory (3)	Satisfactory (2)	Needs Improvement (1)
Panel comment compliance	All panel comments are completely addressed with clear evidence.	Most comments are addressed with minor gaps.	Some comments remain unresolved.	Comments are not addressed.
System revision quality	System revisions improve functionality, usability, or correctness based on feedback.	Most revisions are appropriate.	Revisions are partial or superficial.	No meaningful system revision is evident.
Documentation revision quality	Manuscript revisions are complete, coherent, and properly formatted.	Mostly complete with minor issues.	Several documentation issues remain.	Major documentation issues remain.

Criteria	Excellent (4)	Very Satisfactory (3)	Satisfactory (2)	Needs Improvement (1)
Revision matrix and evidence	Revision matrix is complete, specific, and supported by screenshots/ documents.	Mostly complete with minor gaps.	Incomplete or vague matrix.	No credible revision evidence.
Readiness for routing/deployment	Output is ready for routing, deployment, and final clearance.	Almost ready with minor requirements remaining.	Several requirements remain.	Not ready for routing or deployment.



MIDTERM EXAMINATION

Week: 12 | Time Allotment: 3 Hours

Midterm Examination

The Midterm Examination serves as the comprehensive assessment of the students' progress during the Midterm Period. Instead of a traditional written examination, each capstone group shall present the revised manuscript, updated system, revision records, and compliance documents developed throughout Weeks 7 to 11. The examination evaluates the group's ability to integrate adviser and panel recommendations, demonstrate system improvements, maintain complete documentation, and prepare institutional compliance requirements.

The examination is performance-based and measures both the technical quality of the capstone project and the students' readiness for the succeeding stages of the course.

Examination Components

Each research group shall present and submit the following:

1. Revised Capstone Manuscript

Present the latest version of the manuscript incorporating all approved revisions from adviser consultations, mock defense, and final defense activities.

2. Updated System

Demonstrate the improved version of the developed system, highlighting the implemented revisions, additional features, resolved issues, and overall functionality.

3. Updated Revision Matrix

Present the complete Revision Matrix showing all recommendations received, corresponding actions taken, supporting evidence, and implementation status.

4. Compliance Package

Submit the required compliance documents, which may include:

- Consultation Records
- Panel Recommendation Sheets
- Updated Appendices
- Routing Documents



- Other institutional requirements specified by the course instructor

References

Institutional References

Philippine State College of Aeronautics. (2025). *IT411: Aviation Capstone Project 2 course syllabus* (Version 2, Revision 1, 1st semester, S.Y. 2025–2026). Institute of Computer Studies.

Philippine State College of Aeronautics. (2013). *PhilSCA student research handbook: AY 2013–2016*.

Style Guide

American Psychological Association. (2020). **Publication manual of the American Psychological Association** (7th ed.). American Psychological Association. <https://doi.org/10.1037/0000165-000> (American Psychological Association)

Examination Procedures

Each research group shall:

1. Submit the required documents before the scheduled examination.
2. Present the revised manuscript and updated system within the allotted time.
3. Explain the major revisions implemented after adviser consultations and defense activities.
4. Respond to clarification questions from the instructor.
5. Submit all required compliance documents for verification.

Expected Outputs

Each group shall submit:

1. Final Revised Manuscript (Midterm Version)
2. Updated System Prototype
3. Completed Revision Matrix
4. Compliance Package

Assessment Method

Rubric G – Presentation of Revised Documentation and System

Criteria	Weight
Revised Capstone Manuscript (completeness, organization, incorporation of adviser and panel comments, formatting, and technical accuracy)	30%
Updated System (functionality, improvements, stability, and alignment with research objectives)	30%
Revision Matrix (completeness, documentation of recommendations, actions taken, supporting evidence, and revision status)	20%
Compliance Package (appendices, consultation records, routing documents, and other institutional requirements)	10%
Presentation and Professionalism (organization, communication skills, teamwork, preparedness, and ability to explain revisions)	10%
TOTAL	100%



Performance Descriptors

Rating	Description
Excellent (4)	Output is complete, accurate, professionally presented, and exceeds expectations.
Good (3)	Output is complete with only minor revisions needed.
Fair (2)	Output is acceptable but requires several revisions and improvements.
Needs Improvement (1)	Output is incomplete, poorly organized, or does not meet the required standards.